

HaemoTrack

Concept Development Report



Table of Contents

Table of Contents	1
1. Concept Description	2
2. Technical Development	3
2.1 Prototyping Task 1: Form Factor Validation	3
2.2 Prototyping Task 2: Cannula Efficacy	4
2.2.1 Is negative pressure or a pump required to draw a sample?	4
2.2.2 Will the test reagent be able to reach the bloodstream?.....	5
2.2.3 How will fluid resuscitation through the cannula affect samples drawn?	5
2.3 Prototyping Task 3: Expected Data and Data Presentation	5
3. Research and Development Plan	7
3.1 R&D Milestone 1: Benchtop and Simulated Use Testing	7
3.2 R&D Milestone 2: Acute Animal Model.....	7
3.3 R&D Milestone 3: Proof-Of-Concept Clinical Trial (Surgical)	7
3.4 R&D Milestone 4: User Testing	7
3.5 R&D Milestone 5: Pivotal Clinical Trial (Surgical)	7
3.6 R&D Milestone 6: Pivotal Clinical Trial (Emergency).....	7
4. Regulatory Assessment.....	8
5. IP Assessment.....	9
5.1 IP Landscape Search and Freedom to Operate (FTO).....	9
5.1 Patentable Features (notes).....	9
5.3 IP Protection Strategy	9
6. Design inputs and Risk Assessment	10
Appendix A	11
Appendix B	12
Appendix C	13
References.....	15

1. Concept Description

The HaemoTrack follows a form factor of a bedside photometer. The HaemoTrack would be a device that is connected to an existing intravenous (IV) cannula bifurcation/trifurcation when there has been a suspected blunt force trauma for detection of internal haemorrhage (IH). The IV cannula would attach to the device where cartridges containing a reagent would be automatically loaded. The HaemoTrack would assess the blood sample via a colorimetric method based on reflectance photometry (1). Each cartridge would draw a small defined volume of blood from the cannula's capillary, mix with the reagent, and then analyse the sample to produce a measure within 30-60 seconds. Each cartridge would be single use and once analysed the device would remove the analysed cartridge into a disposal component before reloading and analysing the next cartridge. The HaemoTrack would allow for continuous analysis which would calculate a bleed rate based on the level of constituents analysed over time from multiple samples. Common blood constituents that can change in the presence of an internal haemorrhage include haemoglobin levels, haematocrit, prothrombin time, platelet count, and fibrin degradation products (2). Note that these constituents can be analysed with small quantities of blood (<1mL). In the case of the HaemoTrack, a partial blood count is performed, hence decreased haemoglobin levels and haematocrit are the two parameters that will be measured to detect an IH.

Emergency doctors (and other emergency department clinicians) would be key users of this device and would apply this once a patient is admitted. Due to its small and modular design (bedside or attachable to an IV pole), the HaemoTrack is easily confined and portable, thereby limiting its impact on the workflow of clinicians. The automated cartridge loading, and analysis would help reduce the time to diagnostic imaging and well as ease of use. The addition of a display screen to produce the numerical results obtained from the analysis would also allow a simple user interface requiring minimal training. The main body of the HaemoTrack would be a one-off purchase, with the cartridges being the only consumables to be purchased at a low cost.

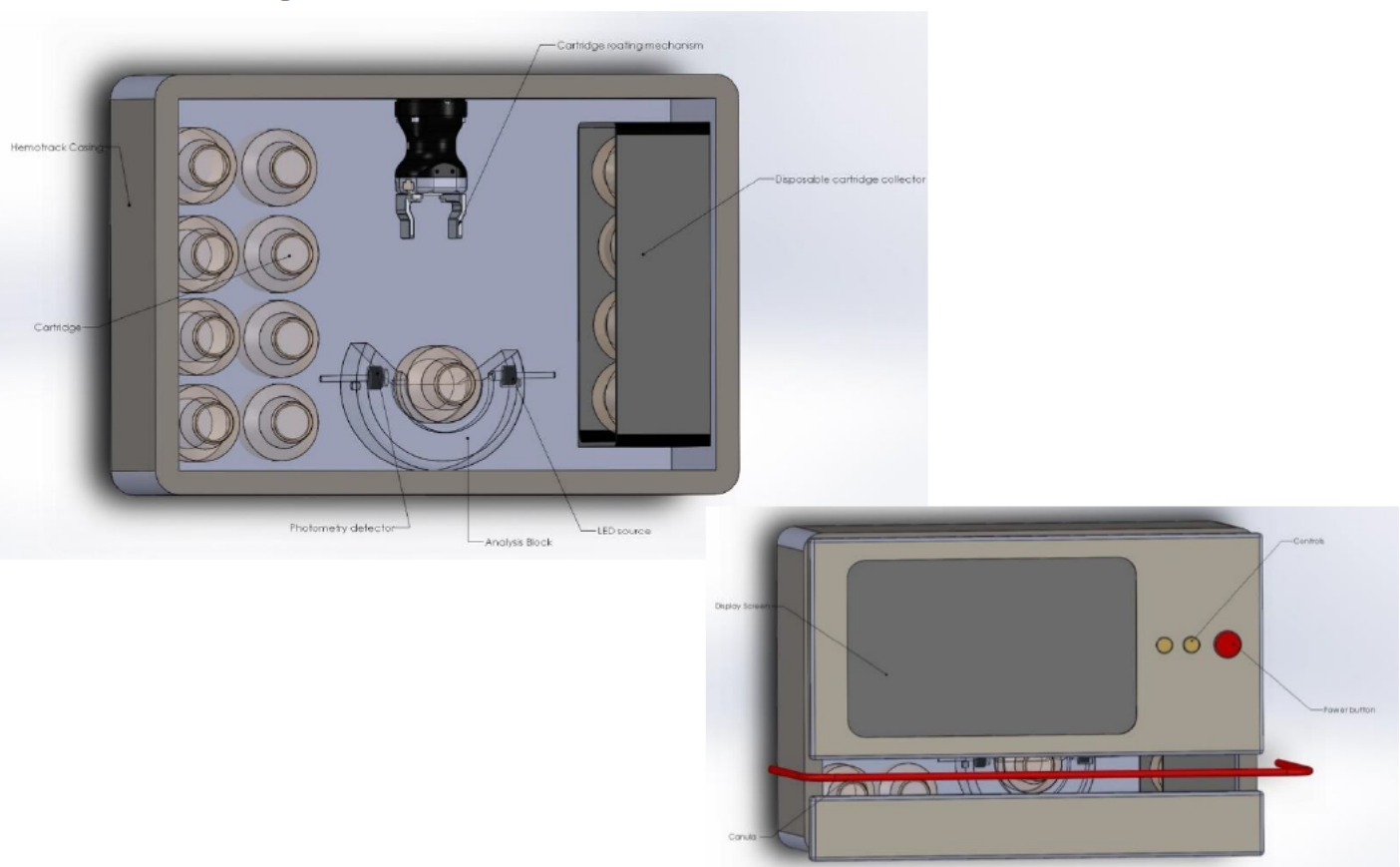


Figure 1.1. Concept Sketch of the HaemoTrack created via Solidworks. (Important to note, currently not shown in model, a clamp is to be attached to the back of the device).

2. Technical Development

The HaemoTrack is intended to be used for monitoring blood constituents for early detection of haemorrhage severity. The top technical risks identified for this medical device are:

1. Can we achieve a sampling frequency for blood analysis that will obtain useful blood measures within a 10-minute time frame to provide useful clinical information?
2. Can we make the device small enough to be viable in predicted use environments?
3. Does the device have high enough sensitivity to reliably output the likelihood of an internal haemorrhage?

These risks have been chosen to be most important due to their potential impact on device usefulness, clinician adoption, and patient health. Risk 1 plays the largest role, and if it is found to be insurmountable will cause the termination of the project. All prototyping to date has some input on addressing this technical risk. Risk 2 addresses ergonomic complications of adding a clinical step in treating critically ill haemorrhaging patients, and how we can reduce or eliminate work-flow obstacles to promote adoption of a potentially life-saving device. Risk 3 focuses on eliminating false negatives, as incorrect diagnosis of the patient may lead to the incorrect treatment and intervention having critical impact.

Thus far the most significant technical development decision has been to focus on the minimum viable product; changing the HaemoTrack to be an automatic blood sampling device attached to an existing cannula (Figure 2.1). The patch may be investigated in the future, but a cannula-based product will accelerate the process to market and can be used to springboard future patch iterations.



Figure 2.1: Models of the HaemoTrack device. First combining and automating a benchtop model, then minimising and attaching to a cannula (minimum viable product). Patch concepts may be explored after-market launch and success of the MVP.

The total prototyping plan for each device stage is summarised in [Appendix A](#). Due to current time, budget, and expertise constraints, many of these prototypes will not be built or tested within the scope of this unit but would be crucial for future development. Our first three prototyping tasks which can be accomplished concurrently are form factor validation (addressing risk 1 and 2), cannula attachment research (risk 2, 3) and researching expected data and possible presentation (risk 1, 3).

2.1 Prototyping Task 1: Form Factor Validation

The HaemoTrack patch concept, follows the condensation of multiple blood tests into a small, mobile device. However, existing designs which test blood focus on the collection of blood and measuring via an external device (1). Hence, the HaemoTrack's patch is a novel approach for performing these tests. It is recognised, that the use of these current blood test devices can be utilised as a proof of concept and the first prototype for the HaemoTrack. For a variety of blood tests, the collection of blood is completed through the use of a cannula (3). This is a common-place medical device utilised in emergency, to general surgery application (3). Due to the high instance of cannula use, it can be effectively utilised to collect and assess blood constituents outside of the body (3).

Blood samples drawn from venous blood is more representative of the physiology than an interstitial finger prick. Also, utilising the cannula already existing in the hospital admission workflow should decrease interruptions to the rapidity of patient care, fit cleanly into the existing workflow and cause minimal additional stress to the patient. (4). Furthermore, without comprehensive validation interviews a Patch concept may not be desired by clinicians in a hospital environment. To validate these claims, further validation interviews surrounding the form factor of the HaemoTrack needs to be completed. Validation interview questions surrounding the form factor can be seen in [Appendix B](#). Additionally, utilising a cannula can help navigate the lagging technological development surrounding the

condensation and combination of tests into a small environment. Finally, utilising a cannula design is low risk for the patient and clinician. This is due to limiting the number of invasive devices entering a patient and limits the need for clinicians to incorporate new devices within their workflow.

To validate the cannulation form factor as an effective substitute for the patch design, and direct future development, a number of validation practices must be completed. This includes: 1) validation interviews with clinicians to validate the value in a novel device that performs these tests, and also confirms their desired form factor; 2) utilise cannula bench top tests as an initial proof of concept to measure constituent within a serum; 3) confirm technical device designs, including the utilisation of a cartridge which collects serum for analysis, disposable components of the device, and how it will connect to existing cannulas within workflow.

2.2 Prototyping Task 2: Cannula Efficacy

The cannula-attachment for the MVP form factor must be well prototyped to investigate possible design changes and danger points for this model. To test questions associated with this risk we created a simple colour-coded flow loop to mimic the vein, cannula, and cannula attachments (Figure 2.2).



Figure 2.2: Equipment for flow loop

2.2.1 Is negative pressure or a pump required to draw a sample?

An open tube to the sample was connected via the cannula analogue to the vein with red blood (Figure 2.3). Horizontal, vertical against gravity and vertical with gravity were tested.

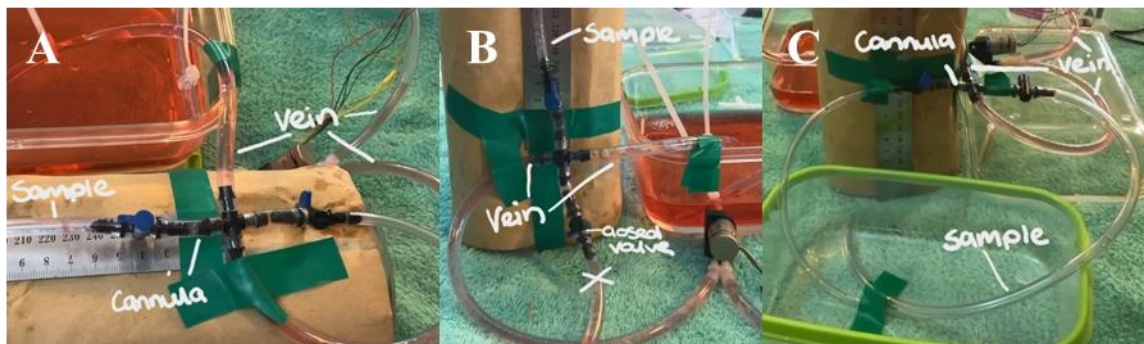


Figure 2.3: Sample draw pressure set up A) horizontal, B) vertical up, and C) vertical down.

The sample had to be extracted with negative pressure in the horizontal and vertical states. The sample flowed freely when the end was 10cm below the vein. Gravity or some negative pressure will be necessary to draw samples. If both sample terminals were opened, proximal venous flow ceased, therefore we must be conscious of the volume of blood drawn for given venous flow rates.

2.2.2 Will the test reagent be able to reach the bloodstream?

A sample of blood was drawn from the vein through the cannula into a syringe with blue test reagent (mimicking the Haemotrack) and monitored for any backflow of contaminated blood (purple) (Figure 2.4). If any purple reached the vein the test would fail.

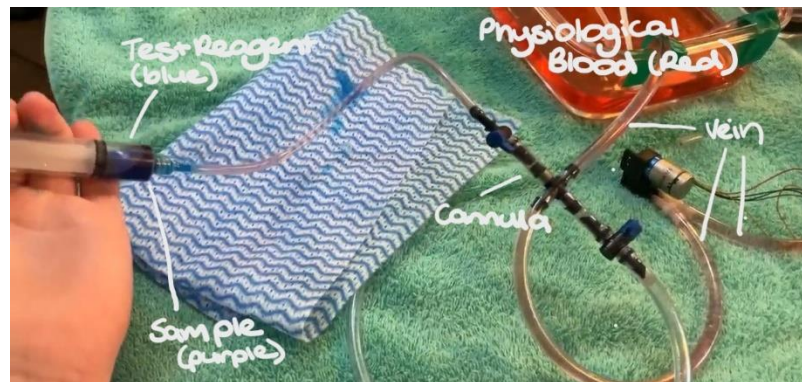


Figure 2.4: Test reagent contamination test set up.

There was minimal movement back toward the vein for both the 3mm and 5mm diameter sample tubes. If any pressure was applied contaminated blood flowed freely toward the vein. A system of valves may be necessary to prevent contamination. Safety is increased with the length of tubing.

2.2.3 How will fluid resuscitation through the cannula affect samples drawn?

Saline (blue) was connected to the vein via the cannula (Figure 2.5). Resuscitated blood turned purple. The HaemoTrack sample must contain physiologically relevant blood measures, therefore our sample must be only red, containing no blue.

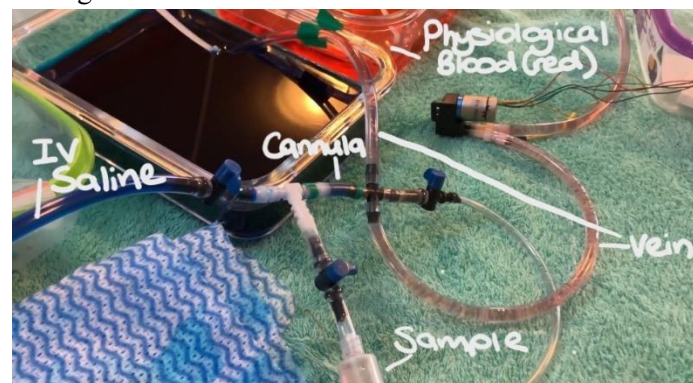


Figure 2.5: Effect of fluid resuscitation on Haemotrack sample.

Valves were necessary to ensure that samples tested were representative of physiological blood (red) and not resuscitated blood (purple). It was also necessary to flush the needle of resuscitation fluid to prevent mixing. Necessity of needle flushing will add downstream design challenges to drawing fresh samples of blood for each time point. Programming the system to account for a known volume of saline from the cannula and tubing may help to address this effect.

2.3 Prototyping Task 3: Expected Data and Data Presentation

Due to the nature of the proposed device, a physical model is not currently feasible. Despite this limitation, our MVP prototype involves building a desktop calculation algorithm that estimates the flow rate of blood loss given the tested variables of either haematocrit (Hct) or haemoglobin levels (Hb). This algorithm will ultimately be used to inform clinicians of the presence of a haemorrhage, and to drive their clinical decision-making.

Average healthy values of adult patients from the literature will be used as the control variables in our calculation (5). These can be found in Table 2.1. Future prototypes will be updated with clinical data to account for pre-existing blood conditions in patients that may affect device performance.

Table 2.1: Average range of healthy adult values for blood tests conducted by Haemotrack.

Patient Group	Healthy Haematocrit (Hct, %)	Healthy Haemoglobin Levels (g/dL)
Male	40 – 54	13.2 – 16.6
Female	36 – 48	11.6 – 15

There are several conditions that readings outside of these healthy values could imply. For the purposes of our MVP prototype, an assumption will be made that through both compensatory physiological responses and IV fluid resuscitation, blood loss causes a decrease in haematocrit. This relationship between blood loss, Hct, and Hb is established in previous studies with a high sensitivity (2,6), but future studies, testing, and validation interviews will aim to further confirm this relationship. An estimation for blood loss flow rate (given by volume-versus-time) will also form another aim in future prototypes as we further understand the mathematical relationships between these parameters of interest.

For our MVP prototype, the following data values will be analysed and displayed on the device as shown in the plot in Figure 2.6. Current devices can provide Hb measurements within 30 – 60 seconds, so a key assumption is that our device can also perform at this frequency.

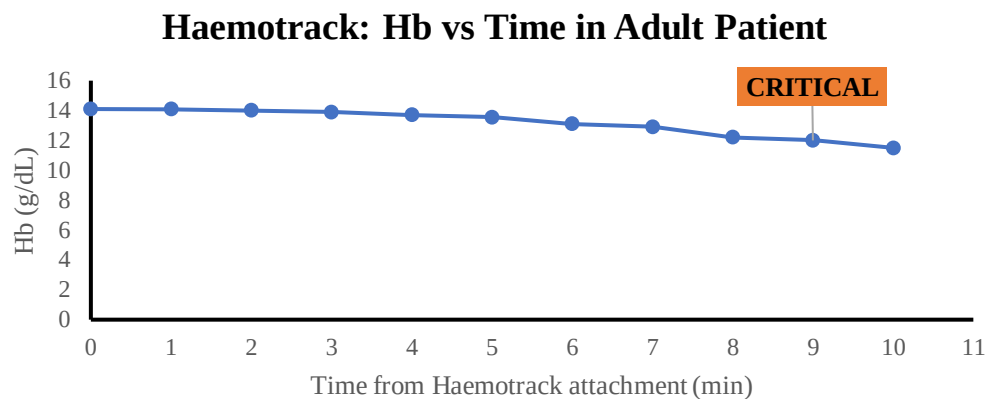


Figure 2.6: Plot of a trauma patient's haemoglobin levels (g/dL) 10 minutes after attaching the Haemotrack.

The plot in Figure 2.6 demonstrates a simulated progression of a trauma patient's haemoglobin levels over an initial 10-minute period right after attachment of the Haemotrack. As indicated at $t = 9$ minutes, the device has detected a high likelihood of haemorrhage. This value in our prototype assumes that a suitable cutoff value to detect severe haemorrhage is at a haemoglobin level of 12g/dL for males and 10g/dL for females, based on a study conducted by Figueiredo et. al.(7). The device would have a user-friendly interface that indicates this information in a concise and efficient matter for clinicians' use. Further establishing this cutoff variable for a definitive haemorrhage indication forms another part of our validation through further testing downstream.

The proposed calculation algorithm requiring further validation would consist of the following stages:

Measured Hct or Hb (time = t) $\rightarrow$ Estimated Blood Volume, EBV (time = t)
 $\rightarrow$ EBV Lost [(EBV at time $t+1$) – (EBV at time t)] divided by t over this period.

Future prototypes will aim to predict the patients' time to mortality if left untreated. The predictive function of this device will be driven by both a curve-fitting feature to the measured constituent values relative to time, and by feeding the device regularly updated clinical information. However, for preliminary studies, we will assess bleeding rate through direct measurement.

3. Research and Development Plan

An initial Gantt chart for the research and development plan is included in Figure 3.1.

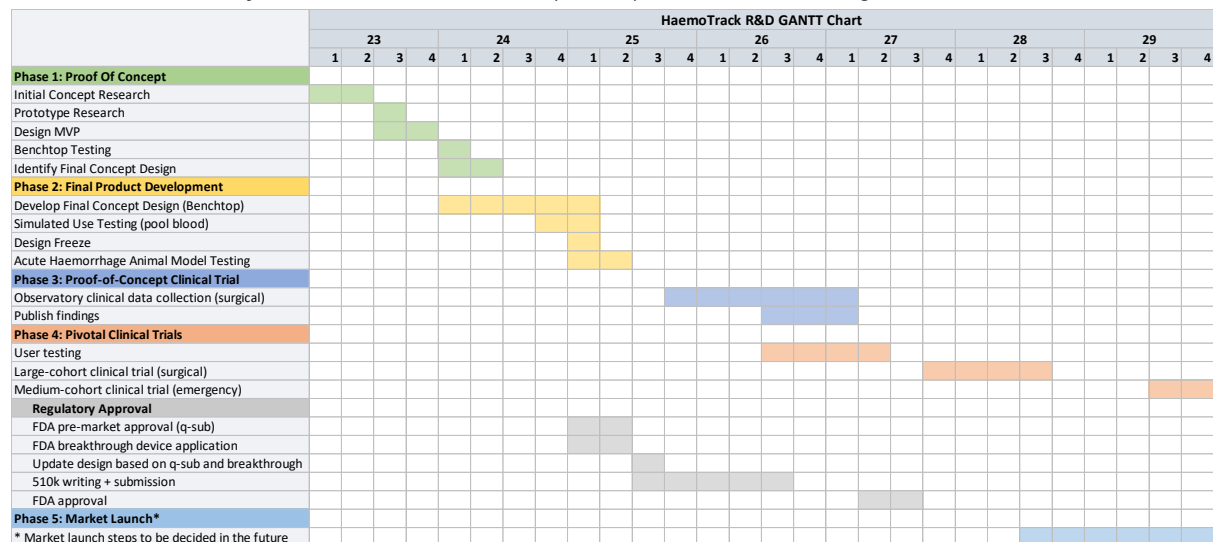


Figure 3.1: Gantt chart for research and development timeline

3.1 R&D Milestone 1: Benchtop and Simulated Use Testing

The Haemotrack will be initially prototyped, iterated, and developed in sections on the benchtop (Appendix A). Simulated use tests will be completed using the final design on a pool of animal blood with increasing saline dilution to confirm that the device can detect changes in haemoglobin concentration over time.

3.2 R&D Milestone 2: Acute Animal Model

Next, an acute animal model will be used to confirm device safety and performance for initial regulatory approval data. Rats are a suitable small animal model for modelling haemorrhage (8). This model will also test efficacy amongst the complex physiological response.

3.3 R&D Milestone 3: Proof-Of-Concept Clinical Trial (Surgical)

A surgical proof-of-concept clinical trial will be conducted. Collecting data and pursuing approval in the more controlled surgical environment will accelerate regulatory approval of our device and help to demonstrate efficacy for marketing and reimbursement. It will require a clinical trial notification (CTN). Patient blood data will be gathered during surgery whilst clinicians perform their usual tasks. This data will not be provided for clinical decisions but used for retrospective correlation with recorded blood measures, volumes and clinical and cost outcome, and published (Section 2.3).

3.4 R&D Milestone 4: User Testing

Whilst waiting on final regulatory approval, user testing and iteration for the Haemotrack workflow and data presentation will be completed.

3.5 R&D Milestone 5: Pivotal Clinical Trial (Surgical)

A large-cohort pivotal clinical trial using the Haemotrack diagnostically will be conducted to get approval for use in the surgical environment.

3.6 R&D Milestone 6: Pivotal Clinical Trial (Emergency)

Based on the success of the surgical clinical trial, a medium-cohort clinical trial may be later conducted for patients in the emergency department to pursue regulatory approval and reimbursement for use in the emergency environment.

4. Regulatory Assessment

The regulatory pathway for FDA approval will follow an initial application for a premarket approval/Q-Sub, and Breakthrough device designation. The premarket approval process can help confirm the selection of our approval pathway, and confirm the regulatory classification of our device (9). While the breakthrough device designation is designed for medical devices that address life threatening diseases and issues, this allows for improved development times through continuous interactions with FDA experts to gain market approval effectively and efficiently (10).

The Haemotrack will collect and perform several haematology tests, which include a haemoglobin blood count. Assessing current FDA regulatory codes, the Haemotrack can be considered a Class II medical device under 21CFR 864, haematology and pathology devices (11). A number of devices currently existing on the market for haematology measures that have already been approved, for example Glycosylated Haemoglobin Assay (K120199). However, both devices have differing indications and intended use to the Haemotrack. This may result in the Haemotrack undergoing a De Novo pathway due to its novel nature, and lack of predicate devices. However, in combinations with the premarket approval, and breakthrough device designation, the FDA can provide guidance if a 510K pathway may become applicable due to the possible approval of a predicate device. This can heavily reduce the required general and special controls to address risks required in a De Novo pathway associated to novel medical devices. Through gaining FDA approval, this can be leveraged to gain approval in other countries including TGA approval, and CE mark.

5. IP Assessment

5.1 IP Landscape Search and Freedom to Operate (FTO)

A Google Patent search related to the device and similar technologies used the following keywords, yielding 300 results.

((TI = "Haemoglobin levels") OR (AB = "Hemoglobin levels") OR (CL = "Hemoglobin levels") OR (TI = "Hematocrit") OR (AB = "Hematocrit") OR (CL = "Hematocrit"))) ((TI = "Photometre") OR (AB = "Photometer") OR (CL = "Photo meter")) OR ((TI = "Reflectance photometry") OR (AB = "Reflectance photometry") OR (CL = "ReflectancePhotometry"))).

The patent landscape surrounding the analysis of haemoglobin concentration and related technologies are diverse and consists of various approaches and design features (Appendix C). Notably, while some patents overlap with Haemotrack, our device has many distinguishing features in comparison. There are a few active patents to be accounted for, as they may have potential implications for Hemotrack market entry and operational strategies. For instance, the US patents, US20110263031A1 and US9651564B2 use photometry to determine the Hemoglobin concentration, with one being abandoned and the other active. CN104931441A and JP2010243373A, patents from China and Japan explore different light absorption mechanisms to determine hemoglobin concentration, but their status remain uncertain or pending. The Korean patent, KR20190114591A is currently pending and describes a similar cartridge design. CN107782670A, now expired, uses a similar cartridge fixing mechanism to Hemotrack. Finally, the active Australian patent, AU2017378148B2, a similar method for multiple liquid concentration determination to Haemotrack's continuous HCT measurements. Despite the overlaps, Haemotrack unique features offer a distinct stance compared to the existing patents. Moreover, with several patents either abandoned, expired or of uncertain status, there's a opportunity to move forward with the development of Haemotrack without infringing on existing patent rights.

5.1 Patentable Features (Notes)

The proposed invention encompasses a novel continuous blood analysis system, designed to greatly enhance diagnostic efficiency and patient care. The cornerstone of this system is its ability to extract a precise amount of blood from a canula piggyback setup, ensuring meticulous volume control for analysis. Once extracted, the system aliquots a defined quantity of blood into individual cartridges for further analysis. This streamlined approach not only expedites the diagnostic process, but also alleviates the need for recurrent clinician intervention, thus eliminating the necessity for multiple blood draws. Complementing this system is an advanced cartridge rotation mechanism. This state-of-the-art component quickly positions a fresh cartridge into the Haemotrack analysis block prior to assessment. Then, this mechanism ensures that after every reading, used cartridges are appropriately discarded and a new cartridge is swiftly placed in the analysis block for subsequent analysis. Notably, if the cartridge exchange mechanism can be completed quickly, this ensures the analysis is completed within mere minutes.

5.3 IP Protection Strategy

Firstly, Haemotrack will apply for utility and design patents for the technology and its appearance. The name "Haemotrack" and associated branding will be trademarked. Unique technical designs will be protected; and software code, manuals, and marketing materials will be copyrighted ensuring thorough protection against infringements.

6. Design inputs and Risk Assessment

*Attached separately.

Appendix A

Milestone 1: Miniaturised 'benchtop'

- Identify critical blood tests
- Miniaturise each blood test (components, cartridge)
- Assess test speed
- How can tests be run in parallel, drawn from same sample?
- Power requirements (mains/battery)
- Can it identify changes (testing; pool blood (healthy), diluted blood (diseased), animal model)
- Identify minimum possible blood volume

Milestone 2: Minimum Viable Product 'Cannula connection- bedside' (physical prototype)

- Collection (automatic/manual)
- Sterility
- Disposal
 - Lifetime (battery)
 - Lifetime (number of tests)
- Automatic test frequency (do we need to collect blood each time we want a result)
- Automatic blood draw (interference or existing cannulas function?- stop drug delivery to draw sample)
- Volume of blood (sample and dead space)
- Pressure draw on blood
- Effect of saline/drug delivery on blood draw (pink into vein= saline, blue vein, is drawn sample blue or purple?)
- How long the pauses are between stop delivery and blood draw
- Things to buy: dual/tri lumen catheter, bags of fluid, blood-like fluid (viscosity), syringes (pull blood), cannulas, tubing, testing device to measure pressure

Milestone 3: Data processing and presentation

- Ideal mode of presentation; single value, present raw data instantly, present figure/graph, page summary
- Ideal result frequency; every 'x' minutes, continuous, update as generated
- Comparative clinical data- research
 - Established methods of presentation
 - Existing benchmarks (healthy vs diseased vs dead)
- Data storage/transmission; raw, complete or processed?
- Presentation; inbuilt screen, wireless transmission (form factor dependent- size)

*R&D questions:

- Type of test (HCT/Hb or PT)
- Type of technology (Ultrasound- too complicated, flow cytometry- too long, reasonable options- biosensor for HCT, photometry)

Appendix B

Validation Interview Qs

Who will we be interviewing.

1. What are the current diagnostic measures for an internal hemorrhage?
 - a. What are their advantages and disadvantages?
2. Which blood tests are important to you?
3. (If not mentioned) How important are the aforementioned blood test results?
[[Tests mentioned Q2, Hb, HCT, FBP, PT]]
 - a. Are they bleeding (y/n)?
 - b. How fast is the patient bleeding?
 - c. Time to bleed out?
 - d. Additional information
4. Which form factor would you prefer? (What is an ideal size of a product to no influence or effect current work flows?)
 - a. Would work for ambulance?
 - b. Would work for handover?
 - c. Would work at the bedside?
 - d. Which would fit best with existing workflow? Existing POC tests?
5. How important is it to have the results displayed on the device screen?
 - a. What results displayed can assist in diagnosing an internal hemorrhage?
6. How important is it to have these results recorded in hospital records?/How long do these results need to be kept?
 - a. (May be temporarily stored on the device, or while patient is in-hospital?)

Appendix C

Table 1 evaluates the most similar patents and their impact on the FTO for the device. Patents highlighted in yellow may pose a challenge in terms of FTO due to their strong similarities to the Haemotrack.

Table1. Haemotrack IP Landscape Assessment: Patents related to Photometry and Haemoglobin Analysis

Patent Title	Status	Claim	FTO and Patentability
US20110263031A1: Process, Portable Equipment and Device for In Vitro, One-Step Photometric Determination of Hemoglobin Concentration in a Diluted Blood Sample	Abandoned	1	This patent describes a method to determine haemoglobin concentration using photometry. However, since the application has been abandoned it poses no significant risk to operation.
US9651564B2: Device and method for measuring hemoglobin	Active	1	The claim describes a haemoglobin analyser which uses specific light wavelengths passing through a cuvette containing a blood sample to determine the haemoglobin concentration, also known as absorbance photometry. Whereas Haemotrack relies on reflectance photometry (Colorimetric Analysis) to measure haemoglobin concentration.
CN104931441A: Method and device for quickly detecting hemoglobin	Uncertain	1	The patent describes a method to accurately determine haemoglobin concentration in undiluted, unhemolysed whole blood. A publication was reported in 2021, however no further progress has been made.
JP2010243373A: Analysis method and analysis apparatus in analysis device	Pending	1	The patent describes a microchannel device which uses centrifugal force, and multiple chambers with reagents to analyse haemoglobin concentration. We are not using centrifugal force to analyse our samples.
US10088468B2: Analyte system and method for determining hemoglobin parameters in whole blood	Active	2	This claim describes an optical absorbance system using spectrometer analysis to determine haemoglobin parameters. Haemotrack will be using colorimetric photometry to analyse the blood sample.
BRPI0822570B1: In vitro process for determining hemoglobin concentration in a sample of diluted blood	Uncertain.	1	The patent covers the methodology for calculating the haemoglobin concentration using colorimetric photometry absorbance data. However, a publication was reported in 2021, with no further progress reported since.
KR20190114591A: Micro Cuvette and Manufacturing Method of the same	Pending	1	The claim describes the methodology for micro cuvette design and manufacture method. Although the manufacturing method is the same, Haemotrack differentiates itself with its unique cuvette (cartage) design.
CN107782670A: A kind of cuvette test fixing device	Expired	1	The patent describes a support block with a light source and a detector block, like Haemotrack 's Photometric block. However, the patent has been expired since 2019, and poses no threat to Haemotrack design features.
AU2017378148B2: Simultaneous measurement of multiple analytes of a liquid assay	Active	11	The claim describes the methodology to determine concentrations of more than two liquids using calibration data and simultaneous equations. This uses the same methodology as Haemotrack to calculate continuous concentrations from absorbance data.

References

1. Lott JA, Khabbaza E. Haemoglobin analysis on whole blood by reflectance photometry. *J Autom Chem*. 1985;7(4):197–200.
2. the Traumabase Group, Figueiredo S, Taconet C, Harrois A, Hamada S, Gauss T, et al. How useful are hemoglobin concentration and its variations to predict significant hemorrhage in the early phase of trauma? A multicentric cohort study. *Ann Intensive Care*. 2018 Dec;8(1):76.
3. Lesser FD, Lanham DA, Davis D. Blood sampled from existing peripheral IV cannulae yields results equivalent to venepuncture: a systematic review. *JRSM Open*. 2020 May;11(5):205427041989481.
4. Yang ZW, Yang SH, Chen L, Qu J, Zhu J, Tang Z. Comparison of blood counts in venous, fingertip and arterial blood and their measurement variation: Comparison of venous, fingertip and arterial blood counts. *Clin Lab Haematol*. 2001 Jun 11;23(3):155–9.
5. Billett HH. Hemoglobin and Hematocrit. In: Walker HK, Hall WD, Hurst JW, editors. *Clinical Methods: The History, Physical, and Laboratory Examinations* [Internet]. 3rd ed. Boston: Butterworths; 1990 [cited 2023 Sep 12]. Available from: <http://www.ncbi.nlm.nih.gov/books/NBK259/>
6. Oman KS. Use of hematocrit changes as an indicator of blood loss in adult trauma patients who receive intravenous fluids. *J Emerg Nurs*. 1995 Oct 1;21(5):395–400.
7. Figueiredo S, Taconet C, Harrois A, Hamada S, Gauss T, Raux M, et al. How useful are hemoglobin concentration and its variations to predict significant hemorrhage in the early phase of trauma? A multicentric cohort study. *Ann Intensive Care*. 2018 Jul 6;8(1):76.
8. Fülöp A, Turóczy Z, Garbaisz D, Harsányi L, Szijártó A. Experimental Models of Hemorrhagic Shock: A Review. *Eur Surg Res*. 2013 Apr 23;50(2):57–70.
9. FDA. Requests for Feedback and Meetings for Medical Device Submissions: The Q-Submission Program [Internet]. Available from: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/requests-feedback-and-meetings-medical-device-submissions-q-submission-program>
10. FDA. Breakthrough Devices Program [Internet]. Available from: <https://www.fda.gov/medical-devices/how-study-and-market-your-device/breakthrough-devices-program>
11. CFR - Code of Federal Regulations Title 21 [Internet]. [cited 2023 May 20]. Available from: <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfCFR/CFRSearch.cfm?fr=890.1615>